EE2021 -
Materials Science for Electrical Engineers

Reciprocal Lattice

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What is reciprocal space?

- Reciprocal domain (frequency) is used to represent time domain signals (Recall from Signals and Systems course)
- Similarly waves in Solids (whether vibrations or electron waves) in real-space can be also represented in Reciprocal space
- Wave vector $k = 2\pi/\lambda$ has the dimensions L^{-1}
- Credited to Ewald whose work in the 1920s laid the groundwork for this concept
- Really useful concept in explaining X-ray diffraction and electronic structure (why Si is not used to make LEDs)

Reciprocal lattice (1)

- Consider a set of points R constituting a Bravais Lattice and a plane wave e^{ikr}. For a general wave vector k, the plane wave will not have any periodicity of the Bravais lattice.
- But for certain wave vectors G, $e^{iG(r+R)}=e^{iGr}$ for all r and R provided $e^{iGR}=1$
- G is the set of all wave vectors that yield plane waves with the periodicity of a given Bravais Lattice
- Given a direct (in real space) lattice of points R, a point G is a point in the reciprocal lattice, iff $e^{iGR}=1$ for all points R
- A reciprocal lattice is defined wrf to a particular Bravais lattice

How to define reciprocal space?

- In real space, the primitive unit vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$
- We will define the unit vectors $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{b}_3$ in reciprocal space, which relate to real space vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ as

$$\mathbf{b_1} = \frac{2\pi\,\mathbf{a_2}\times\mathbf{a_3}}{\mathbf{a_1}\cdot(\mathbf{a_2}\times\mathbf{a_3})}$$

$$\mathbf{b_2} = \frac{2\pi\,\mathbf{a_3}\times\mathbf{a_1}}{\mathbf{a_1}\cdot(\mathbf{a_2}\times\mathbf{a_3})}$$

$$\mathbf{b_3} = \frac{2\pi\,\mathbf{a_1}\times\mathbf{a_2}}{\mathbf{a_1}\cdot(\mathbf{a_2}\times\mathbf{a_3})}$$

$$\mathbf{a_i}\cdot\mathbf{b_j} = 2\pi\delta_{ij}$$

where δ_{ij} is the Kronecker delta.[2]

- Does the above make reciprocal lattice a Bravais lattice?

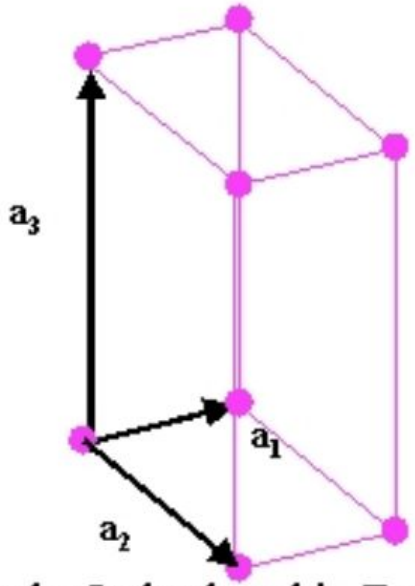
Reciprocal lattice is a Bravais Lattice

- Let us write an arbitrary point in reciprocal space as $G = m_1b_1 + m_2b_2 + m_3b_3$.
- We can also write similarly for a point R in direct lattice, $R = n_1a_1 + n_2a_2 + n_3a_3$ where n_1, n_2 and n_3 are integers. But for certain wave vectors G, $e^{iG(r+R)}=e^{iGr}$ for all r and R provided $e^{iGR}=1$
- Given a direct (in real space) lattice of points R, G is a point in the reciprocal lattice, iff $e^{iGR}=1$ for all points in R

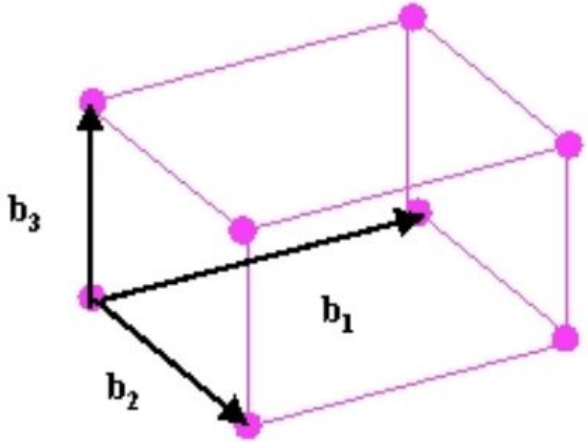
$$e^{i\mathbf{G}\cdot\mathbf{R}} = e^{i(m_1\mathbf{b_1}+m_2\mathbf{b_2}+m_3\mathbf{b_3})\cdot(n_1\mathbf{a_1}+n_2\mathbf{a_2}+n_3\mathbf{a_3})} = e^{2\pi i(n_1m_1+n_2m_2+n_3m_3)}$$

- It can be unity only if m_1, m_2 and m_3 are integers. This shows all points G constitutes a Bravais Lattice

Constructing Reciprocal lattice



Simple Orthorhombic Bravais Lattice
with $a_3 > a_2 > a_1$

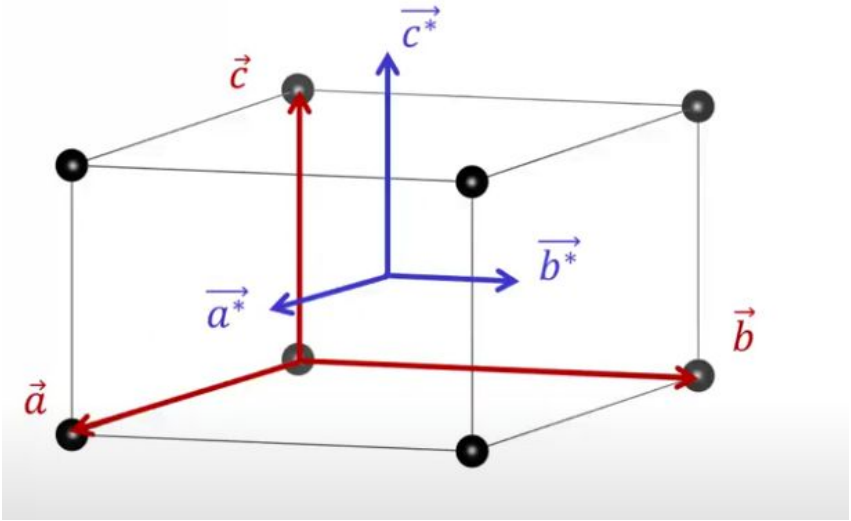


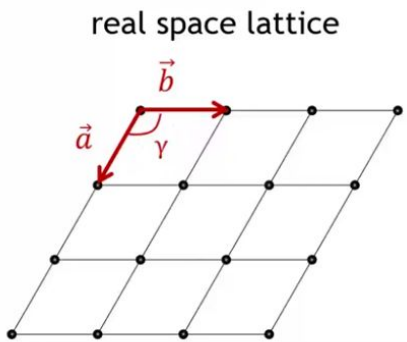
Reciprocal Lattice
Note: $b_1 > b_2 > b_3$

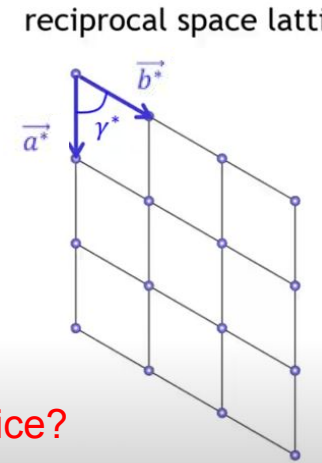
Long lengths in real space imply short lengths in reciprocal space and vice versa.

What can you say about their directions?

Direction of reciprocal lattice vector







What is c here for a 2D lattice?

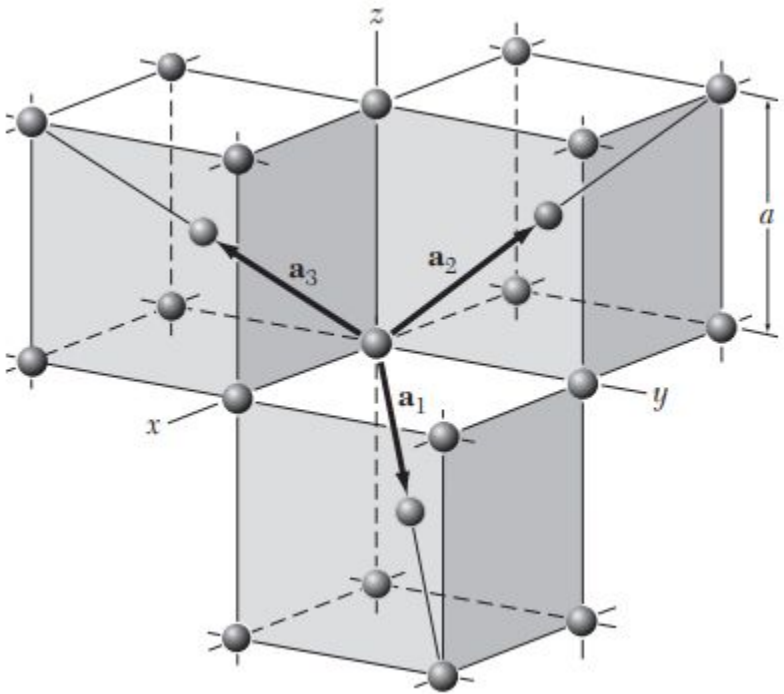
$$\vec{b} \cdot \vec{a^*} = 0 \qquad \vec{a} \cdot \vec{b^*} = 0 \qquad \vec{a} \cdot \vec{c^*} = 0$$

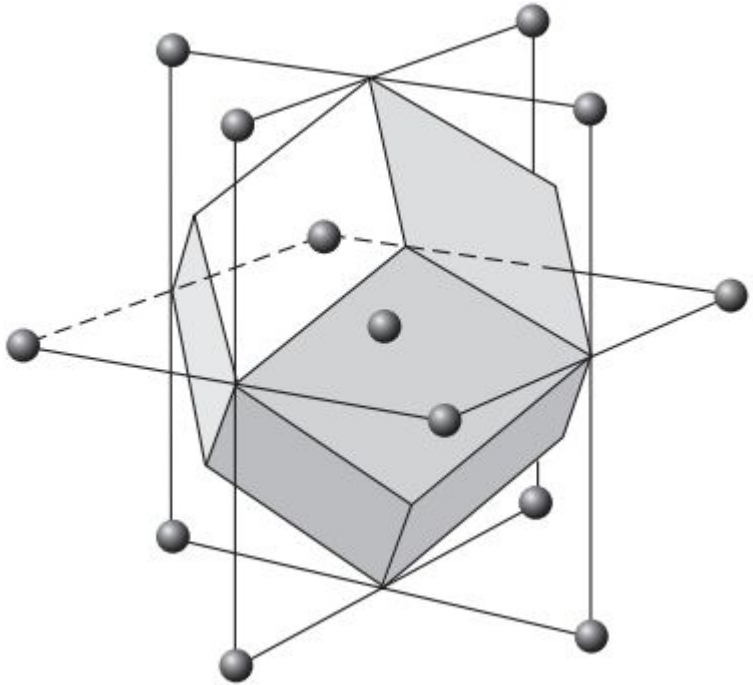
$$\vec{c} \cdot \vec{a^*} = 0 \qquad \vec{c} \cdot \vec{b^*} = 0 \qquad \vec{b} \cdot \vec{c^*} = 0$$

b* is perpendicular to the ac plane/face of unit cell

a&a*/ b&b* / c&c* are parallel only if the real space lattice vectors are orthogonal

Reciprocal lattices of BCC

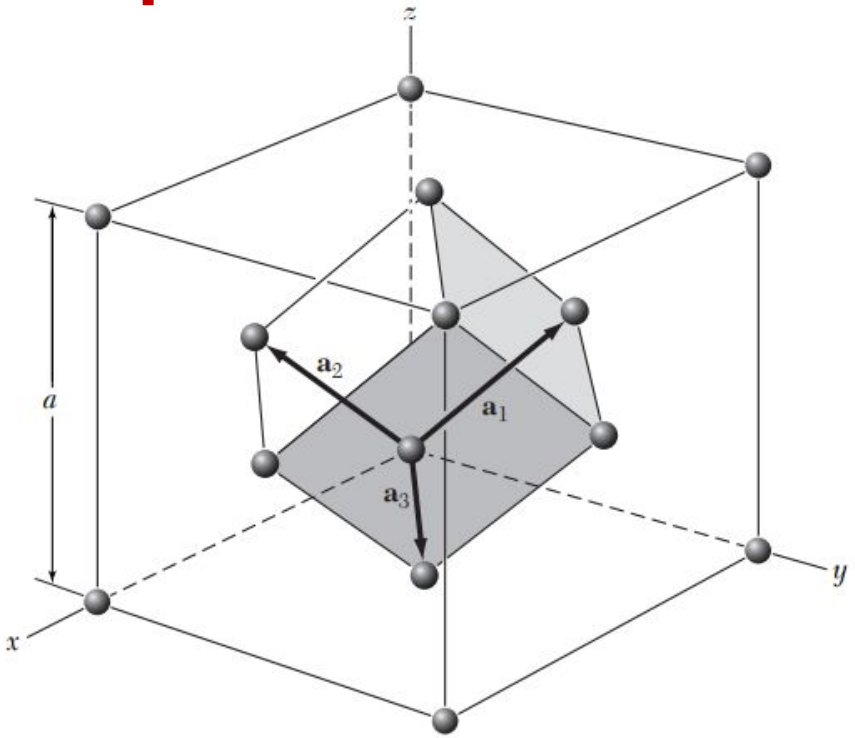


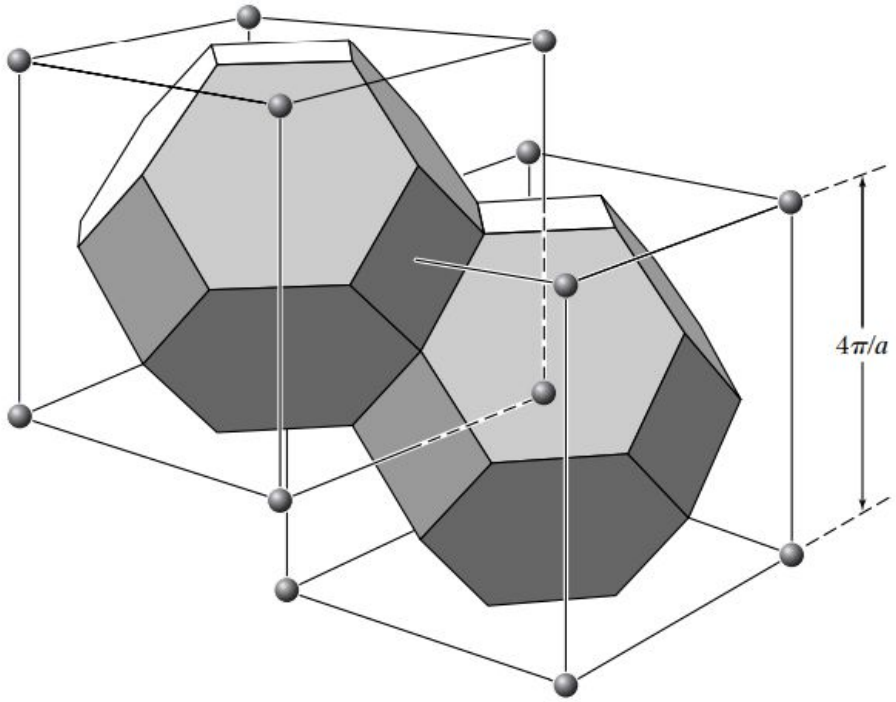


$$\mathbf{a}_1 = \tfrac{1}{2}a(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}) \; ; \quad \mathbf{a}_2 = \tfrac{1}{2}a(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}) \; ; \quad \mathbf{a}_3 = \tfrac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}})$$

$$\mathbf{b}_1 = (2\pi/a)(\hat{\mathbf{y}} + \hat{\mathbf{z}}) \; ; \qquad \mathbf{b}_2 = (2\pi/a)(\hat{\mathbf{x}} + \hat{\mathbf{z}}) \; ; \qquad \mathbf{b}_3 = (2\pi/a)(\hat{\mathbf{x}} + \hat{\mathbf{y}})$$

Reciprocal lattices of FCC





$$\mathbf{a}_1 = \frac{1}{2}a(\hat{\mathbf{y}} + \hat{\mathbf{z}}) \; ; \quad \mathbf{a}_2 = \frac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{z}}) \; ; \quad \mathbf{a}_3 = \frac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{y}})$$

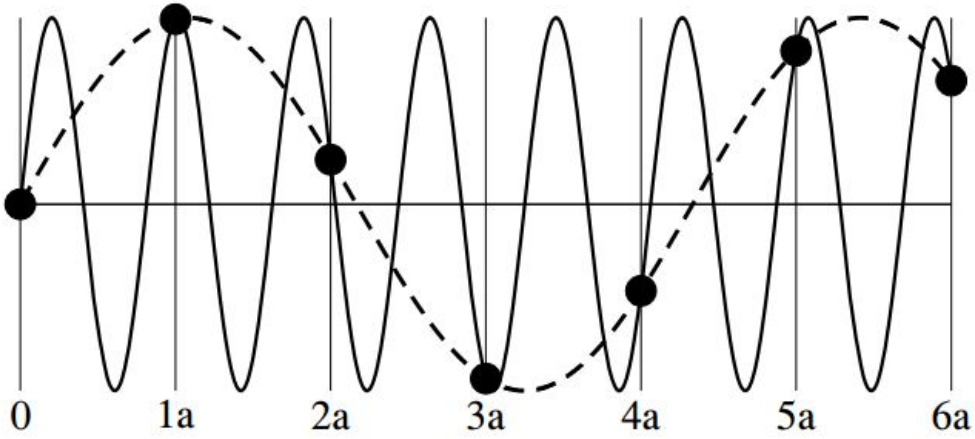
$$\mathbf{b}_1 = (2\pi/a)(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}) \; ; \qquad \mathbf{b}_2 = (2\pi/a)(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}) \; ;$$

$$\mathbf{b}_3 = (2\pi/a)(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}) \; .$$

Aliasing

A wavevector k describes the same wave as the wavevector k + 2π/a.

$e^{ir(k+2\pi/a)}=e^{ikr}$, only at r=na where $e^{i2\pi r/a}=1$



The dashed curve has wavevector k whereas the solid curve has wavevector k + 2π/a. These two waves have the same value (solid dots) at the location of the lattice points x_n = na, but disagree between lattice points. If the physical wave is ONLY defined at these lattice points the two waves are fully equivalent.

Similar aliasing happens when sampling rate in time domain is lower than Nyquist rate of signals

Reciprocal lattice as a Fourier transform

- Reciprocal lattice can be thought as being a Fourier transform of the direct lattice
- Direct lattice is given by R_n = an. We can put a delta function at each lattice points and write the density of lattice points as

$$\rho(r) = \sum_n \delta(r - an).$$

Fourier transforming this function gives[4]

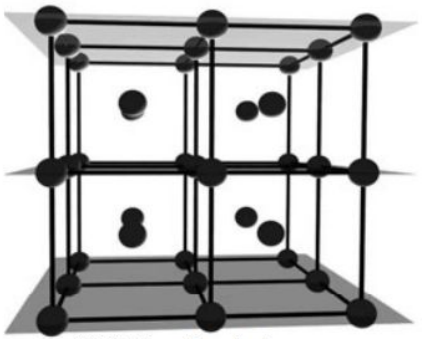
$$\mathcal{F}[\rho(r)] = \int dr e^{ikr} \rho(r) = \sum_n \int dr e^{ikr} \delta(r - an) = \sum_n e^{ikan}$$
$$= \frac{2\pi}{|a|} \sum_m \delta(k - 2\pi m/a).$$

- e^{ikan} is unity if k = 2πm/a, i.e., if k is a point on the reciprocal lattice.
- Points are periodic in k space with periodicity 2π/a

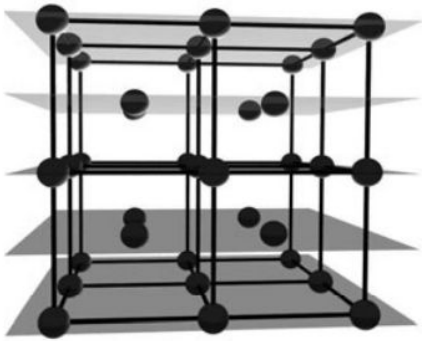
Vector normal to any lattice plane

- A lattice plane is a plane containing at least three non-collinear points of a lattice
- A family of parallel lattice planes is an infinite set of equally separated parallel lattice planes which taken together contain all points of the lattice.
- Let (hkl) be any plane not passing through origin, with intercepts (x_i,y_i, and z_i,) on the respective axes with unit cell of dimensions (a,b,c)

$$h : k : l :: \frac{a}{x_i} : \frac{b}{y_i} : \frac{c}{z_i}$$



(010) family of planes
(not all lattice points included)

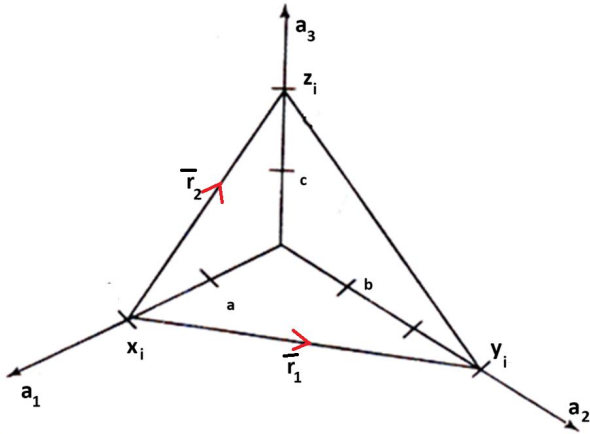


(020) family of lattice planes

- It can be proved that G = hb_1 + kb_2 + lb_3 the reciprocal lattice vector is normal to the above real lattice plane with miller index (hkl)

$$\vec{r_1} = \left(\frac{y_i}{b}\right)\vec{a_2} - \left(\frac{x_i}{a}\right)\vec{a_1}$$

$$\vec{r_2} = \left(\frac{z_i}{c}\right)\vec{a_3} - \left(\frac{x_i}{a}\right)\vec{a_1}$$



$$\vec{r_1} \times \vec{r_2} = \left(\frac{y_i z_i}{bc}\right)(\vec{a_2} \times \vec{a_3}) + \left(\frac{x_i z_i}{ac}\right)(\vec{a_3} \times \vec{a_1}) + \left(\frac{x_i y_i}{ab}\right)(\vec{a_1} \times \vec{a_2})$$

$$= \left(\frac{a y_i z_i}{abc}\right)(\vec{a_2} \times \vec{a_3}) + \left(\frac{b x_i z_i}{abc}\right)(\vec{a_3} \times \vec{a_1}) + \left(\frac{c x_i y_i}{abc}\right)(\vec{a_1} \times \vec{a_2})$$

$$= \left(\frac{x_i y_i z_i}{2\pi}\right)\left[\left(\frac{a}{x_i}\right)\vec{b_1} + \left(\frac{b}{y_i}\right)\vec{b_2} + \left(\frac{c}{z_i}\right)\vec{b_3}\right]$$

$$= A\left[h\vec{b_1} + k\vec{b_2} + l\vec{b_3}\right]$$

$$\mathbf{b_1} = \frac{2\pi\,\mathbf{a_2} \times \mathbf{a_3}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})}$$

$$\mathbf{b_2} = \frac{2\pi\,\mathbf{a_3} \times \mathbf{a_1}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})}$$

$$\mathbf{b_3} = \frac{2\pi\,\mathbf{a_1} \times \mathbf{a_2}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})}$$

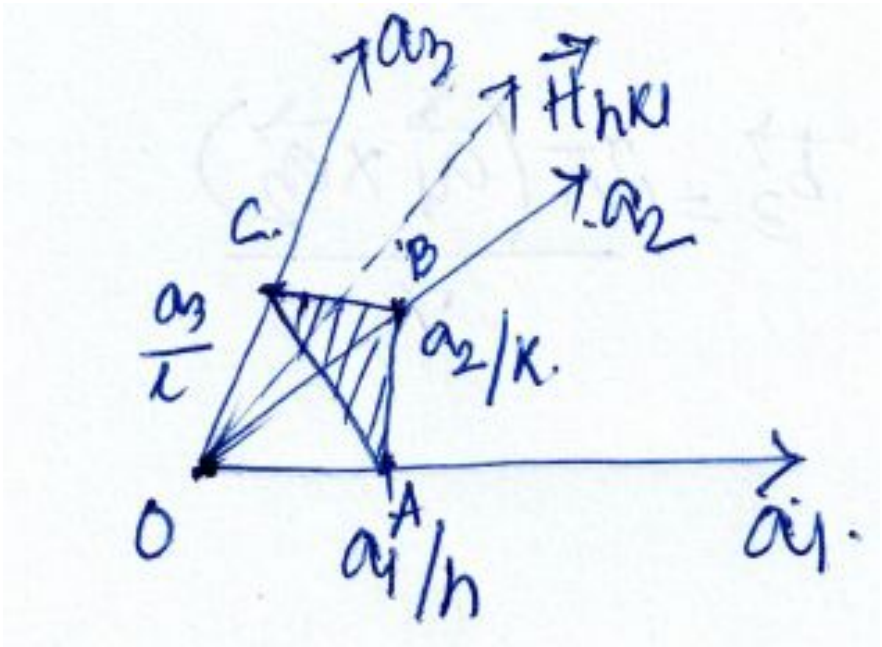
- $G = hb_1 + kb_2 + lb_3$ the reciprocal lattice vector is normal to the above real lattice plane with miller index (hkl)

Vector normal to any lattice plane

- Two successive planes of indices (hkl) make intercepts on the crystal axes of na/h, nb/k, nc/l and (n + 1)a/h, (n + 1)b/k, (n + 1)c/l, respectively, where n is an integer
- One can make one plane going through origin. Then equation of other plane becomes hx/a + ky/b + lz/c =1
- Then the perpendicular distance (d_{min}) between origin and the plane can be written as

$$\frac{1}{d_{min}^2} = \left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 + \left(\frac{l}{c}\right)^2$$

Relation between real and reciprocal lattice



$\vec{H}_{hkl} = h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3$

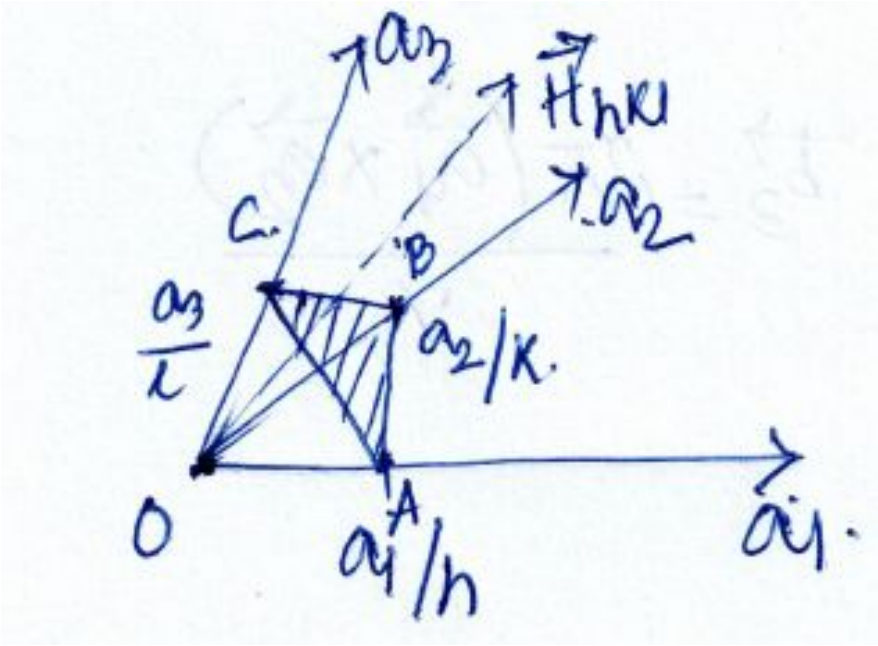
Reciprocal lattice vectors.

** (h, k, l) plane in real lattice

∴ Intercepts along a_1, a_2 & a_3 are $\frac{a_1}{h}$, $\frac{a_2}{k}$ and $\frac{a_3}{l}$

Currently $\vec{H}_{hkl}$ is an arbitrary drawn vector. (D)

Relation between real and reciprocal lattice



$$\vec{OA} = \frac{\vec{a_1}}{h} \qquad \vec{OB} = \frac{\vec{a_2}}{k} \qquad \vec{OC} = \frac{\vec{a_3}}{l}$$

$$\vec{OA} + \vec{AB} = \vec{OB}$$

$$\frac{\vec{a_1}}{h} + \vec{AB} = \frac{\vec{a_2}}{k}.$$

$$\boxed{\vec{AB} = \frac{\vec{a_2}}{k} - \frac{\vec{a_1}}{h}}$$

Relation between real and reciprocal lattice

$$\vec{b}_i \cdot \vec{a}_j = 2\pi\delta_{ij}$$

$$\vec{H}_{hkl} \cdot \vec{AB} = (h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3) \cdot \left(\frac{\vec{a}_2}{k} - \frac{\vec{a}_1}{h}\right)$$

$$= (h)\left(\frac{-1}{h}\right)2\pi + (k)\left(\frac{1}{k}\right) \cdot 2\pi$$

$$= -2\pi + 2\pi = 0$$

$\therefore \vec{H}_{hkl}$ is $\perp$ to $\vec{AB}$

Similarly, we can prove $\vec{H}_{hkl}$ is $\perp$ to $\vec{BC}$ as well as $\vec{CA}$

$\therefore \vec{H}_{hkl}$ is $\perp$ to plane defined by ABC or (h, k, l).

Relation between real and reciprocal lattice

Unit vector along $\vec{H}_{hkl}$ is $\hat{n} = \dfrac{\vec{H}_{hkl}}{|\vec{H}_{hkl}|}$.

Spacing between (h,k,l) planes is distance from origin to (h,k,l) plane

$$\therefore d_{hkl} = \hat{n} \cdot \frac{\vec{a}_1}{h} = \frac{\vec{a}_1}{h} \cdot \frac{(h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3)}{|\vec{H}_{hkl}|} = \frac{\frac{\vec{a}_1}{h} \cdot h\vec{b}_1}{|\vec{H}_{hkl}|} = \frac{2\pi}{|\vec{H}_{hkl}|}$$

$$\therefore |\vec{H}_{hkl}| = \frac{2\pi}{d_{hkl}}$$ and $\vec{H}_{hkl}$ is $\perp^r$ to (h,k,l) plane

Lattice planes and Reciprocal lattice

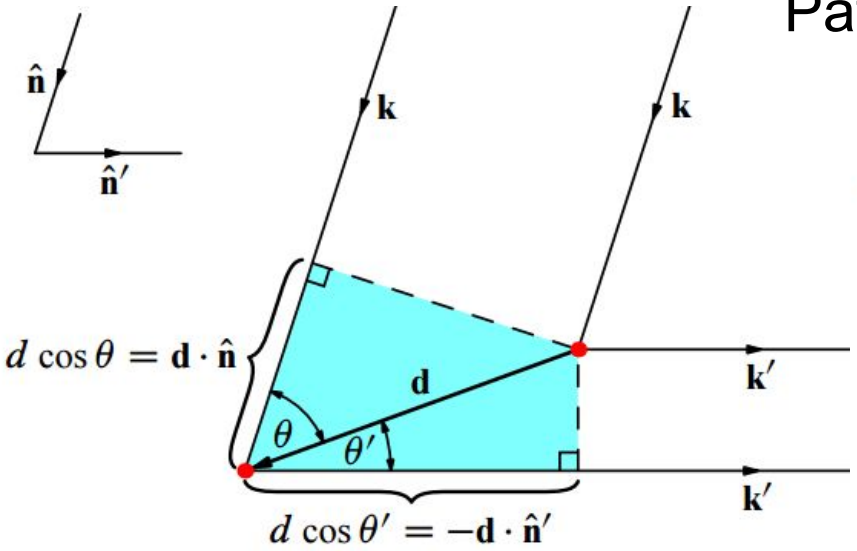
- The spacing between these lattice planes is $d = 2\pi/|G_{min}|$ where G_{min} is the minimum length reciprocal lattice vector in this normal direction

$$d_{(hkl)} = \frac{2\pi}{|\mathbf{G}|} = \frac{2\pi}{\sqrt{h^2|\mathbf{b_1}|^2 + k^2|\mathbf{b_2}|^2 + l^2|\mathbf{b_3}|^2}}$$

- For orthogonal axes and cubic system,

$$d^{cubic}_{(hkl)} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Max von Laue's condition



Path difference: $d\cos\theta + d\cos\theta' = \mathbf{d}\cdot(\hat{\mathbf{n}} - \hat{\mathbf{n}}')$

Condition for constructive interference:

$\mathbf{d}\cdot(\hat{\mathbf{n}} - \hat{\mathbf{n}}') = m\lambda$, for integral m

Multiplying both sides by $2\pi/\lambda$ yield a condition on incident and scattered wave vectors as

$\mathbf{d}\cdot(\mathbf{k} - \mathbf{k}') = 2\pi m$, for integral m

Array of scatters at the sites of Bravais lattice → **d** is bravais lattice vector i.e. **R**

∴Condition for constructive interference: $\mathbf{R}\cdot(\mathbf{k} - \mathbf{k}') = 2\pi m$

for integral m and all Bravais lattice vectors **R**

Laue's condition

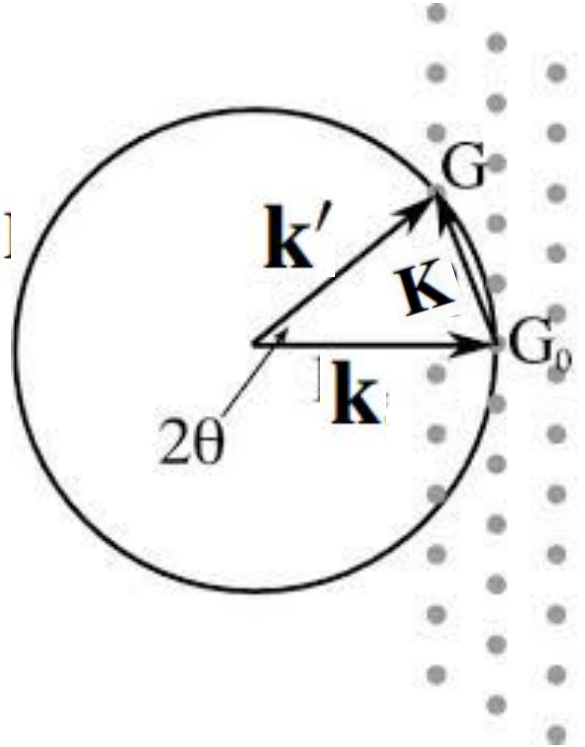
$$\mathbf{R} \cdot (\mathbf{k} - \mathbf{k}') = 2\pi m$$

$$e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{R}} = 1, \quad \text{for all Bravais lattice vectors } \mathbf{R}$$

Comparing the condition with definition of reciprocal lattice vector,

$$\mathbf{K} = \mathbf{k}' - \mathbf{k}$$

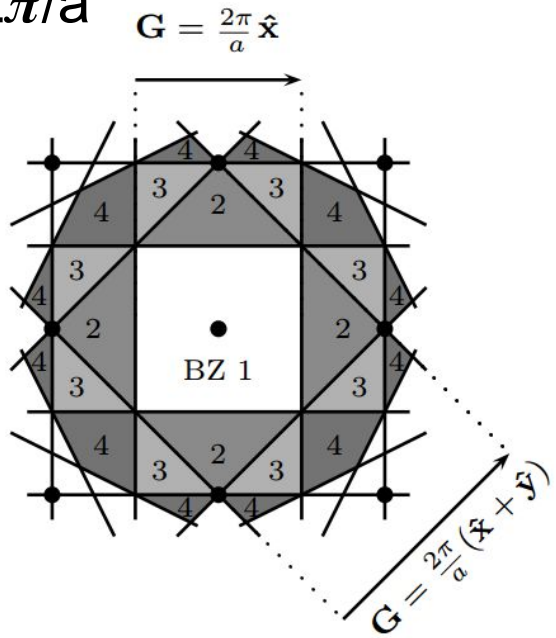
Constructive interference will occur provided that the change in wave vector is a vector of reciprocal lattice



Ewald sphere in reciprocal space

Brillouin zone

- 1st **Brillouin zone** is the Wigner-Seitz primitive cell of the reciprocal lattice
- Since waves are physically equivalent under shifts of the wavevector k by a reciprocal lattice vector $2\pi/a$ (G), we can do all representation within the first Brillouin zone (BZ)
- N^{th} Brillouin zone (BZ) is the region in reciprocal space between n^{th} and $(n-1)^{th}$ **Bragg planes** (perpendicular bisectors to the line joining the origin and any reciprocal lattice points)



Diffraction in Brillouin zone boundary

- Whenever a Bragg plane touches one end of a wave vector, the condition for diffraction is satisfied
- Diffraction occurs whenever a wave vector touches BZ boundary

